

ChordShot: Bridging Vision and Music with AI-Driven, Context-Aware Music Generation tool

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Abstract—Vision-to-music generation is an emerging field with broad applications in film scoring and video production. We introduce ChordShot, an AI system that automatically composes 30-second music clips from images by analyzing visual features and mapping them to musical parameters. It extracts scene and object semantics and dominant colors using pretrained vision models, then applies learned semantic mappings (for example, bright colors map to major keys and lively instrumentation) to specify tempo, mood, and instrumentation. This novel pipeline bridges computer vision and generative music, leveraging MusicGen, a state-of-the-art transformer-based music generator – to synthesize coherent audio conditioned on visual cues. ChordShot’s fully automatic, intuitive framework supports diverse creative applications, from background scoring for multimedia content to interactive art and gaming, alleviating the need for manual composition. Preliminary evaluations indicate that ChordShot’s outputs exhibit strong contextual coherence and emotional alignment with input images, consistent with recent affective vision-music studies.

Index Terms—Music Generation, Scene recognition, Objects analysis, Context based Music

I. INTRODUCTION

The past decade has witnessed transformative advancements in both artificial intelligence (AI) and deep learning, particularly in sequence modeling and audio synthesis. In parallel, computer vision technologies have made remarkable strides in interpreting complex visual scenes, identifying objects, emotions, and aesthetic features. Despite these advances, the integration of computer vision and music generation remains a relatively unexplored domain. This paper introduces **ChordShot**, a novel AI system that bridges this gap by transforming visual inputs into corresponding musical compositions, offering a pioneering approach to visual-to-music translation.

Recent research in affective computing demonstrates that visual elements, including colors, objects, and scene types, are strongly linked to emotional responses (Sartori et al., 2015; Wang et al., 2020). Psychological studies further support the

existence of consistent mappings between color palettes and musical emotions (Palmer et al., 2013). ChordShot builds on these insights by leveraging image analysis techniques to extract dominant visual features—such as objects (e.g., beaches, forests, urban settings), color dominance (warm vs. cool tones), and scene context (nature, architecture, abstract forms)—which are then encoded into semantic representations.

Research Question 1: How effectively can visual features (such as objects, colors, and scene contexts) be translated into musically coherent compositions that are contextually aligned with the emotional tone of the image?

Our evaluation results show that the system’s image-to-music translation pipeline generates compositions that are not only musically coherent but also contextually resonant with the visual input. User feedback and alignment metrics indicate a strong correspondence between the visual stimuli and the emotional tone of the generated music, confirming the effectiveness of the approach.

The system uses state-of-the-art object detection and color analysis algorithms to create these representations. These semantic features are subsequently fed into MusicGen, a transformer-based model capable of generating rich, stylistically coherent music from conditional inputs. [2] Through this process, ChordShot automatically derives musical attributes such as tempo, mood, and instrumentation, generating 30-second musical clips that are contextually aligned with the input images.

Unlike traditional music generation methods that typically rely on textual or symbolic prompts (Agostinelli et al., 2023), ChordShot innovates by using images as the primary creative input. This visual-to-music pipeline provides a novel interaction paradigm, where visual cues, rather than abstract descriptions, drive the creative process.

Research Question 2: What is the role of dominant visual elements (objects, colors, and scene contexts) in determining the

musical attributes such as tempo, mood, and instrumentation?

The study highlights that objects and colors play a significant role in shaping the musical mood, tempo, and instrumentation. For instance, scenes with natural elements like trees and beaches tend to generate more organic sounds, while urban settings evoke more mechanical rhythms. Color analysis further influences mood-related parameters, such as assigning warm colors (like red or orange) to faster tempos and cooler tones (like blue) to slower, more ambient music.

This paper outlines the ChordShot framework, detailing its image analysis module, feature encoding strategies, and the heuristic mappings that translate visual features into musical parameters. Furthermore, we present a comprehensive evaluation based on user studies and feature alignment metrics. Our results demonstrate that ChordShot produces music that users perceive as both emotionally resonant and thematically consistent with the input image. These findings suggest that ChordShot offers compelling applications in personalized media creation, adaptive gaming soundtracks, and immersive art installations, heralding a new frontier in AI-powered, context-aware music generation.

II. BACKGROUND AND RELATED WORKS

AI-driven music generation has advanced significantly, particularly with deep learning techniques in sequence modeling and generative models. Notable systems like MusicVAE (Roberts et al., 2018) and OpenAI’s Jukebox (Dhariwal et al., 2020) have enabled the creation of complex, high-fidelity musical compositions. However, these systems primarily rely on text or symbolic data inputs, with limited exploration of cross-modal generation from visual stimuli.

Experiments in affective computing have supported close links between visual and affective responses and hence visual-music mapping. For example, Cowie et al. (2001) emphasized the importance of visual influences on auditory response, and pioneer research such as Revill and Hevner (1936) examined colour-music matches, which are developed further within recent studies like Lee and Narayanan (2005). Tools such as Riffusion (Forsgren and Martiros, 2022) have developed real-time music synthesis but are oriented towards sound patterns instead of visual input. Conversely, ChordShot fills the gap between music generation and computer vision. Through the analysis of image features—e.g., objects, color composition, and scenes—ChordShot synthesizes contextually relevant music, guided by color-music theories (Palmer et al., 2013) and emotional responses of visual stimuli. Integrating state-of-the-art image analysis with generative music models such as MusicGen, ChordShot provides a new paradigm where images generate the process of music production, unveiling new potential in multimedia content creation.

III. METHODOLOGY

The ChordShot system presents a new cross-modal approach that maps visual semantics extracted from static images into corresponding music pieces. The suggested architecture is built

upon three fundamental stages: image analysis, semantic feature encoding, and conditional music generation, each playing an important role in making the generated music both reflect the emotional and thematic content of the input visual.

The first phase, Image Analysis, detects high-level information from the input image utilizing a set of computer vision methodologies optimized for musical mapping. YOLOv8 is employed for object detection, detecting main objects like trees, oceans, cars, or human bodies. Concurrently, a specialized scene classification network assigns the general environmental context to labels like natural areas, urban environments, or internal spaces. In order to represent the perceptual emotional content of the image, dominant color palettes are obtained by means of k-means clustering performed in the HSV (Hue-Saturation-Value) color space. The color information is essential in determining the emotional quality of the music generated, as determined through known color warmth and affective state associations.

After feature extraction, the second phase, Semantic Feature Encoding, sums up the visual features into a structured semantic vector. This vector represents object presence, scene type, and color temperature and forms a small but dense encoding of the image’s semantic meaning. ChordShot establishes empirically determined correspondences between such visual features and musical parameters informed by psychological studies of color-emotion correspondence and musical affect. Images dominated by warm tones, such as reds and oranges, paired with expansive scenes, are typically associated with upbeat, major-key music featuring faster tempos and acoustic instruments. Conversely, cooler tones like blues and greens, often found in more isolated or tranquil scenes, tend to be linked with slower tempos, minor keys, and more ambient or atmospheric textures. This mapping ensures that the emotional undertones of the visual content are translated into music.

During the third phase, Conditional Music Generation, the composed semantic vector acts as the condition input for a MusicGen model that has been fine-tuned. MusicGen, a transformer generative architecture, is modified to process structured semantic inputs and generate musically coherent outputs that conform to the visual and emotional context. The conditioning mechanism governs high-level musical elements like key, tempo, instrumentation, and overall mood. ChordShot also provides for tunable hyperparameters to facilitate genre-specific variations, thus providing flexibility to accommodate the generated compositions in various creative contexts or user tastes.

In total, the ChordShot system works by means of a modular and extensible pipeline that evolves from semantic analysis of visual inputs to affectively-aligned controlled musical generation. Not only does this design facilitate high-quality image-to-music translation, but it also facilitates independent improvements at each step, for example, by adding more fine-grained emotional recognition models or future advancements in music generation architectures.

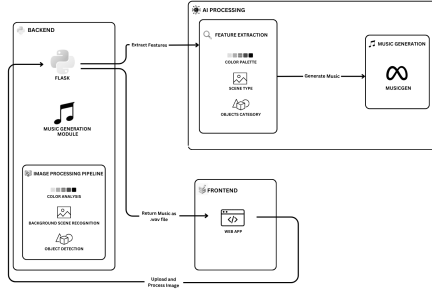


Fig. 1. Visual representation of our methodology.

IV. EXPERIMENTATION

A. Phase 1: Scene Classification

The initial step of our pipeline is the classification of the input image into a background scene category. This classification is a semantic hint for the subsequent music generation process, affecting both the emotional tone and structural characteristics of the resulting audio.

We make use of the *Scene-15 dataset* [10], which consists of grayscale images across 15 unique scene classes, such as *bedroom*, *living room*, *MITforest*, *MITstreet*, and *CALsuburb*. Images were divided into training and test sets using standard benchmark protocols.

In order to well capture both global and local visual features, we employ a hybrid feature extraction method. Local texture features are extracted by densely sampling *DAISY descriptors* [8], and global structure is extracted by *Histogram of Oriented Gradients (HOG)* [3]. DAISY descriptors are clustered by *MiniBatchKMeans* [6] to build a 256-word visual vocabulary, which forms a *Bag-of-Visual-Words (BoVW)* [7] histogram per image. This histogram is concatenated with the global HOG descriptor and L2-normalized to form the final image representation vector.

We compare two classifiers on this representation: an *RBF SVM* [1] and a *Linear SVM* [4] (LinearSVC). The RBF SVM, with hyperparameters $C \approx 48$ and $\gamma \approx 0.687$, has the highest test accuracy of 76.4%, marginally better than the Linear SVM, which had 75.5%.

At deployment, when a user uploads an image via the interface, the trained classifier makes a prediction of the scene category in real time. This predicted label is then fed to the subsequent stage to condition the music generation module so that the audio is contextually consistent with the scene.

B. Dominant Color Extraction

In the second stage of our pipeline, we concentrate on extracting the dominant colors from the input image. This step is important in determining the emotional tone and atmosphere of music that we produce. Color exerts immense influence, not just on the visual appeal but on the emotive experience of sound as well.

1) *How We Extract Colors:* We perform *K-Means clustering* [6], a common machine learning technique, to cluster similar pixel colors according to their RGB values. Here's how it goes:

- **Image Preprocessing:** We read the image through OpenCV and change its color mode from BGR (OpenCV's default) to RGB for uniformity.
- **Reshaping the Image:** The image is reshaped into a 2D array where each row is a single pixel and mapped to its relative RGB color values.
- **Applying K-Means:** We apply K-Means clustering with $k = 4$ to separate all pixels into four dominant color groups. Each cluster is a dominant color in the image.
- **Choosing Dominant Colors:** Out of these four clusters, we select the two most dominant colors, the ones that occupy the largest area of the image.

2) *Color Matching:* When we have the two prominent colors, we pair each with the closest color name from the *CSS3 color set*. If there's a perfect match, we'll use that name. Otherwise, we compute the closest color in RGB space using Euclidean distance.

3) Implementation Overview:

- **Image Handling:** OpenCV [1] is used to read images and convert colors.
- **Clustering:** The *KMeans* module of the *sklearn* library manages the color clustering.
- **Color Naming:** We use the *webcolors* Python library to match CSS3 color names for RGB values.

4) *Output and Results:* For each image that we processed, we did the following:

- Identified the top two most dominant colors.
- Mapped those colors to their closest CSS3 names.
- Stored the resulting color data in a formatted JSON for future use.

5) *Example Output:* In one example image, the dominant colors identified were:

- **Color 1:** Royal Blue
- **Color 2:** Lime Green

These visual color cues were then conveyed to the music generation model to guide the mood and atmosphere of the generated music, causing the music to become more attuned to the visual input.

C. Object Detection with YOLOv8

The third phase of our system pipeline uses YOLOv8 for object detection within the input image. Identifying the objects present in the scene can further enhance the generated music by integrating additional context, such as specific items or activities within the scene. This module provides an understanding of the objects and activities in the environment, which can influence both the mood and thematic elements of the music.

1) *Model Overview*: We employed the **YOLOv8** model [9], a recent and efficient architecture from the YOLO (You Only Look Once) family of networks. Designed for real-time object detection, YOLOv8 demonstrates state-of-the-art accuracy and speed. The model was pre-trained on the **COCO dataset** [5], enabling it to detect over 80 object classes.

2) *Object Detection Methodology*: The object detection pipeline included several key steps:

- We initialized a pre-trained YOLOv8 model using the `ultralytics` library, which provides an intuitive interface for model loading and inference.
- Inference was conducted on input images using a confidence threshold of **0.25**; detections above this threshold were considered valid.
- The output of the model included *bounding boxes* and *class IDs*, which were subsequently mapped to human-readable object names based on the COCO class labels.

3) *Implementation Details*: The model was deployed following these ordered steps:

- 1) **Model Loading**: A pre-trained YOLOv8 model was loaded from a local path using the command `YOLO('yolov8m.pt')`, referencing the medium-sized model variant.
- 2) **Inference**: Input images were processed through the `model()` function.
- 3) **Output Extraction**: Bounding boxes and class IDs were extracted from the inference results.
- 4) **Class Mapping**: The detected class IDs were translated to semantic labels using the COCO object classes.

The final object names were stored as an array of text labels for use in downstream tasks.

4) *Results*: The YOLOv8 model successfully identified and labeled objects in a variety of input scenes. For example, in an image of a living room, the model returned object labels such as "sofa", "person", "television", and "cup". These labels served as an intelligible summary of the visual content of the scene.

5) *Example*: In one test image, the YOLOv8 model detected the following object classes:

- **person**
- **dog**
- **laptop**
- **keyboard**

These detected objects were subsequently passed to other modules within the system pipeline, particularly to influence the contextual music generation process based on the semantic content of the scene.

D. Music Generation using MusicGen-Small

In the final stage of the pipeline, the visual attributes derived from the image, i.e., scene category, dominant color, and objects present, are used to contribute in creating an enjoyable musical piece through the MusicGen-Small model. It serves as a point of music synthesis, where information from the multimodal

conceptualization of the image is transformed into an output of sound.

The system starts by generating a music generation directive that retains affective and semantic content of the image based on a predefined mapping from a salient color found from color analysis to a mood tag and a tempo attribute. For instance, shades of blue might evoke '*smooth electronic beats with gentle transitions*', whereas red would mean '*energetic rhythms with strong percussion*'. Musical output is hence driven by mood of the image.

Scene classification helps identify space and environment setup of a request, such as where the scene might be 'urban,' 'forest' or 'coastal.' It dictates the overall mood and architecture of a piece, determining the musical genre and arrangement to utilize. As an illustration, a 'coastal' scene might be suited to *calm, flowing sound*, while 'urban' environments might require more *rhythmic and energetic forms*. By establishing a context, scene classification helps in developing the mood of the produced music.

Object detection contributes to the level of instrumentation of the music by identifying real objects in the environment. These objects, e.g., 'waves,' 'birds,' or 'cars,' are then mapped to appropriate instruments in accordance with their properties. For example, '*oceanic pads*' can be used to mimic waves, '*flutes*' to represent birds, and '*industrial percussions*' to represent cars. The object grounding of the image enhances the connection between objects detected and aural experience, making the music more contextually engaging as well as more relatable to real life.

The resulting prompt is a descriptive written phrase, e.g.,

"A upbeat track inspired by a beach, featuring yellow tones and synthesizer, piano, guitar, flute. The composition blends ambient layers with soundscapes reflecting the environment, including gentle oceanic textures, chirping melodies or light flutes"

The generated prompt is then sent to facebook/musicgen-small, a transformer conditional audio generation model. The prompt is tokenized using AutoProcessor, properly converted into embeddings, and sent to the MusicGen model. It generates an audio waveform of about 30 seconds, conditioned merely upon the semantic richness of the image-derived prompt.

The resulting raw audio tensor is then saved as a final waveform as .wav file at a 16kHz rate using `torchaudio.save`, completing the generation pipeline. The sound is played back in the frontend interface and also make user downloads possible.

V. CONCLUSION, LIMITATIONS, AND FUTURE WORK

ChordShot presents a new method of crossing over from visual perception to musical creativity. Having analyzed key elements of a photo—predominating coloring, scene type, and objects perceived—ChordShot generates a musical preview which encapsulates the aesthetic and contextual significance of the picture. This preview is employed to guide the MusicGen-Small model in the generation of a short evocative musical

composition. This is a streamlined process of computer vision and AI compositional generation which portends unprecedented degrees of personalized content generation, digital artwork generation, and multimedia storytelling.

Although creative in its undertaking, ChordShot has several limitations. Its visual-to-description mapping is based on pre-defined rules at the moment and may limit the variety and emotional depth of the resulting music. Secondly, the MusicGen-Small model does not process images as such and works with interpreting text input only, which may result in a disconnection of the visual input and the musical output in certain applications. Thirdly, the system only outputs 30-second long snippets at a 16kHz sampling frequency and may thus be inadequate in applications which demand high-fidelity sound or longer musical pieces. Lastly, the processing time to generate music may not always be conducive to real-time interaction with its user interface on limited hardware configurations.

Future evolution of ChordShot aims to overcome these constraints and enhance its functions. One of them is to improve generation of prompts with deeper natural language processing and more image comprehension to generate more contextualized and dynamic prompts. Another direction of interest is to integrate user feedback mechanisms where users can adjust and personalize the produced music to their own taste. Utilization of more powerful and larger models like MusicGen-Medium or switching to a hybrid model may improve sound and be capable of longer and more structured pieces of music. Use of real-time visual-to-music generation and interactive and immersive customer experience is also expected to set higher standards and open more doors to its use in the domains of art, education, and leisure.

REFERENCES

- [1] Gary Bradski. Opencv: Open source computer vision library. *Dr. Dobb's Journal of Software Tools*, 2000.
- [2] Valentin Copet, Alexandre Défossez, Jade Copet, Julien Gertsch, Oriol Nieto, Gabriel Synnaeve, and Mohamed Abdalla. Simple and controllable music generation. *arXiv preprint arXiv:2306.05284*, 2023.
- [3] Navneet Dalal and Bill Triggs. Histograms of oriented gradients for human detection. In *IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 886–893. IEEE, 2005.
- [4] Rong-En Fan, Kai-Wei Chang, Cho-Jui Hsieh, Xiang-Rui Wang, and Chih-Jen Lin. Liblinear: A library for large linear classification. *Journal of machine learning research*, 9:1871–1874, 2008.
- [5] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context. https://doi.org/10.1007/978-3-319-10602-1_48, 2014.
- [6] J. MacQueen. Some methods for classification and analysis of multivariate observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1:281–297, 1967.
- [7] Josef Sivic and Andrew Zisserman. Video google: A text retrieval approach to object matching in videos. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1470–1477. IEEE, 2003.
- [8] Engin Tola, Vincent Lepetit, and Pascal Fua. Daisy: An efficient dense descriptor applied to wide-baseline stereo. <https://doi.org/10.1109/TPAMI.2009.77>, 2010.
- [9] Ultralytics. YOLOv8 : Real-time object detection with improved accuracy and efficiency. <https://arxiv.org/abs/2301.08603>, 2023.
- [10] Ye Xu, Xiaodong Yu, Tian Wang, and Zezhong Xu. Scene-15 dataset. <https://figshare.com/articles/dataset/Scene-15/12103434>, 2020.